

MedRemind — Project Overview

Your Smart Medicine Reminder — a mobile app (with a browser demo) that helps patients and caregivers in Bangladesh never miss a dose.

Quick links

What	Link
Live web demo	https://theobsidianeye-arif-foysal.github.io/Medremind_v2/
Source code (GitHub)	https://github.com/TheObsidianEye-ARIF-FOYSAL/Medremind_v2
User guide (PDF, with screenshots)	https://theobsidianeye-arif-foysal.github.io/Medremind_v2/medremind_user_guide.pdf

The web demo rebuilds and redeploys automatically every time code is pushed to the `main` branch — always reflects the latest version.

What the app does

- **Medicine Cabinet** — a personal list of every medicine the user takes (brand name, generic name, strength, form), pulled from a built-in database of common Bangladeshi brand names.
 - **Dose Groups (Schedules)** — bundle medicines into named times of day (Morning/Afternoon/Evening/Night) with an exact alarm time, optional meal relation (before/after meal), and repeat days.
 - **Full-screen alarms** — ring at the scheduled time even if the screen is off or another app is open, with Dismiss (Taken) / Snooze / Skip actions directly on the alarm screen or its notification.
 - **Home dashboard** — today's doses with a live progress ring, filterable by time of day.
 - **Daily Planner** — full-month calendar with a colour-coded adherence dot per day (all taken / partial / missed / scheduled).
 - **History** — weekly adherence percentage, a monthly heatmap, and a full chronological dose log.
 - **Account** — phone number + password sign-in, verified once at registration via an SMS OTP; profile screen with logout, change password, and unsubscribe.
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How access works

The app is distributed through **BDApps** (Robi/Airtel direct-carrier billing): subscribing costs a small amount per day, billed to the user's mobile operator account, verified by an SMS OTP tied to their phone number.

- **Native mobile app** (Android): full functionality — real OTP/SMS via BDApps, local SQLite storage, real system alarms and notifications.
- **Web demo**: for viewing/reviewing the UI and flows only.
 - Uses [Device Preview](#) so it can be viewed inside a phone-shaped frame in any browser.
 - Local data (medicines, dose groups, dose history) is stored in the browser's IndexedDB instead of a native SQLite file — it works for demoing the full CRUD flow, but each browser/device has its own separate data, and OS-level permissions/alarms/notifications don't apply on web (those calls are safely no-op'd).
 - The subscription/OTP screens work against the real BDApps API, which during testing only accepts a small set of BDApps-whitelisted phone numbers (see "BDApps testing status" below).

Tech stack

Layer	Technology
App	Flutter (Dart), Riverpod for state, go_router for navigation
Local storage	SQLite via <code>sqflite</code> (native) / IndexedDB via <code>sqflite_common_ffi_web</code> (web)
Alarms/notifications	<code>alarm</code> + <code>flutter_local_notifications</code> packages (native only)
Auth/subscription	Custom PHP + SQLite server (<code>server/</code>), talking to the BDApps API
Hosting (web demo)	GitHub Pages, auto-deployed by GitHub Actions on every push to <code>main</code>

Full technical breakdown of the BDApps integration (endpoints, request/response shapes, status codes) is in `server/BDAPPS_INTEGRATION.md`.

BDApps testing status

The app's BDApps application (`APP_138840`, branded "MedRee") is currently approved for **testing only** — BDApps will only process OTP/subscribe/unsubscribe requests for a short list of phone numbers they've whitelisted for this testing phase. Any other number is rejected by BDApps

itself (independent of anything in the app or server code).

Once testing is complete, requesting **Active Production** from BDApps (email support@bdapps.com) opens it up to any real number.

Repository layout

med_remind_v2/	
├─ app/	Flutter app source
├─ server/	PHP + SQLite backend (auth, OTP, subscription)
├─ docs/	
│ └─ report_MR_app/	LaTeX user guide + screenshots (PDF included)
└─ .github/workflows/	CI: builds the web demo and deploys to GitHub Pages

Security note

BDApps credentials are kept in `server/bdapps_config.php`, which is gitignored and never committed — see `server/bdapps_config.example.php` for the template. If you're setting this server up fresh, copy the example file, fill in the real `BDAPPS_APP_ID /` `BDAPPS_APP_PASSWORD`, and deploy `bdapps_config.php` alongside the rest of `server/` on your host.